



University of Colombo School of Computing
SCS 1304 - Problem Solving Strategies and Computation Approaches
Lab Sheet 03

Sequential Search, Binary Search & Fibonacci sequence

You are working as a data analyst in an online shopping company. The company stores thousands of customer order records every day. Some records are arranged randomly, while others are sorted by Order ID.

You are asked to analyze customer orders using search algorithms and identify sales growth patterns using the Fibonacci sequence.

Part 1 – Sequential Search & Binary Search

- A. Imagine the customer orders are stored randomly in a list and are not arranged in any specific order. If you need to find the order with Order ID **ORD4589**, how would you use a **Sequential Search** to locate it? Describe the steps involved.
 - B. Now imagine the customer orders are arranged in ascending order based on the Order ID. How would **Binary Search** help you find the order **ORD4589** more efficiently? Describe the steps involved.
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You are given the following unsorted list of order numbers:

[105, 78, 210, 56, 132, 89, 305, 176, 44, 250]

- 1. Use **Sequential Search** to find the number **176**. How many comparisons are required?
 - 2. Sort the list in ascending order and use **Binary Search** to find the number **176**. How many comparisons are required?
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Part 2 – Scenario: Website User Growth Analysis

A technology startup is monitoring the number of new users joining its mobile application every month. The company notices that the growth pattern follows a **sequence**, where each month's number of new users is the sum of the previous two months.

The startup recorded the following data:

- ★ Month 1 → 1 new user group
- ★ Month 2 → 1 new user group
- ★ Month 3 → 2 new user groups
- ★ Month 4 → 3 new user groups
- ★ Month 5 → 5 new user groups

The company wants to predict future user growth to prepare server capacity, storage, and marketing strategies.

Questions

1. Using the Fibonacci sequence, calculate the number of users for the:
 - I. 6th month
 - II. 8th month
2. If the company wants to estimate the total number of users by the end of the 8th month, how many users should they expect in total?
3. If the growth pattern continues, how many users will there be in the 12th month?
4. Calculate the average monthly growth rate between the 6th and 12th months.
5. Determine the total number of users from the 1st month to the 12th month.
6. Another application follows a different growth sequence where each month's user growth equals the sum of the previous **three months**. Assuming the first three months are 1, 1, 2, calculate the number of users in the 6th month. Compare this value with the previous sequence value of the 6th month.

7. The growth rate is defined as the difference in user count between consecutive months. Identify the month with the highest growth rate between the 1st and 12th months.
 8. Create a simple graph (or explain how you would create one) showing user growth from the 1st to the 12th month. Describe any trends or patterns observed.
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Part 3 – Applying Search Algorithms to Fibonacci Data

You are given the Fibonacci sequence for the first 15 months:

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610

- I. Use **z** to find the position of the number **34**. Describe the process and calculate the number of comparisons required.
- II. Use **Binary Search** to find the position of the number **144** in the sorted Fibonacci list. Describe the process and calculate the number of comparisons required.
- III. Suppose the company wants to determine in which month the number of users reaches **233**. Explain how both:
 - A. Sequential Search
 - B. Binary Search

can be used to find this information. Discuss the advantages and disadvantages of both methods.

- IV. Use **Binary Search** to determine the range of months where the number of users falls between **21 and 233**. Explain the steps involved in performing this search.
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Part 4 - Try the following LeetCode tasks and Submit screenshots of the successfully completed outputs.

[Binary Search, Fibonacci Number, Search Insert Position]